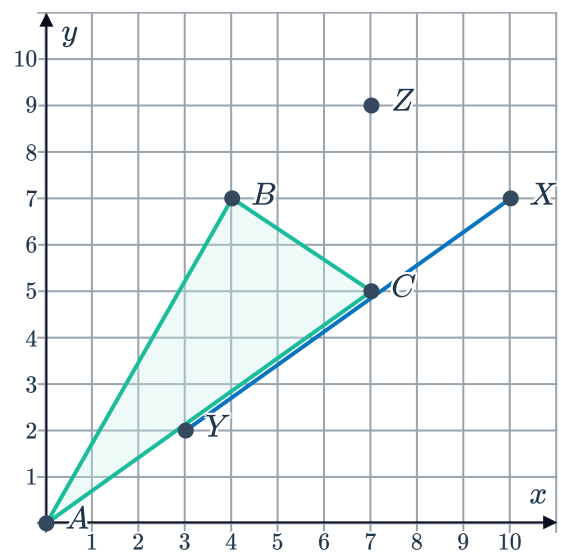
Coordinate methods and triangle congruence

1
- a Translation by 3 units left and 3 units down
- b Z translates to C and Y translates to A
- c Yes

2
- a Translation by 3 units left and 3 units down.
- b Z does not translate to any vertex of $\triangle ABC$ and Y translates to B.
- c No

3



4 No

5 Yes. The triangles are reflections of one another.

6 Yes. The triangles are rotations of one another.

7
- a
  - i $AB = \sqrt{29}$
  - ii

Segment	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	$\overline{XY}$	$\overline{YZ}$	$\overline{ZX}$
Length	$\sqrt{29}$	$\sqrt{13}$	$\sqrt{34}$	$\sqrt{29}$	$\sqrt{13}$	$\sqrt{34}$

  - iii Yes

b

i $AB = \sqrt{65}$

ii

Segment	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	$\overline{XY}$	$\overline{YZ}$	$\overline{ZX}$
Length	$\sqrt{65}$	$\sqrt{13}$	$\sqrt{74}$	$\sqrt{65}$	$\sqrt{13}$	$\sqrt{74}$

iii Yes

c

i $AB = \sqrt{89}$

ii

Segment	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	$\overline{XY}$	$\overline{YZ}$	$\overline{ZX}$
Length	$\sqrt{89}$	$\sqrt{10}$	$\sqrt{113}$	$\sqrt{89}$	$\sqrt{10}$	$\sqrt{113}$

iii Yes

d

i $AB = \sqrt{53}$

ii

Segment	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	$\overline{XY}$	$\overline{YZ}$	$\overline{ZX}$
Length	$\sqrt{53}$	$\sqrt{26}$	$\sqrt{85}$	$\sqrt{68}$	$\sqrt{37}$	$\sqrt{113}$

iii No